Fix the Loss-Making Locations: Diagnose First, Then Treat Each Site

Tong Wu · abrightlab 2,000-Location Challenge · Written recommendation + Python diagnostic model (simulated data)

My answer in one line: a loss-making location is not one problem but six different ones. Build a per-location profit and root-cause model on Peazy + finance data, give every losing site a diagnosis and a matching fix, and re-organise vendors by geography. In a simulated test on two badly set-up metros, this cut losing sites from 200 to 14 and lifted gross margin from 0% to about 29%, while two untouched control metros kept drifting down.

1. Problem framing and assumptions

The company is a general contractor: the customer pays it one monthly price for many sites, and it pays local cleaning vendors to do the work. A location loses money when **customer price < vendor cost**. I frame the goal as **location-level gross margin**, not vendor price, because the loss can come from either side of that equation, or from the way the network is laid out.

- Price is set per site per month for a contracted scope and frequency; vendors are paid per visit or per month.
- Labour is the main cost and tracks local wage x hours; travel is a real cost that vendors build into their quotes.
- Quality cannot drop: a fix that saves money but lowers inspection scores or retention is not a fix.
- Peazy check-in/out (geofence), photos and inspections are reliable enough to measure what was actually delivered.

2. Data I would need, and why

Already in Peazy	Likely outside Peazy (finance, sales, HR)
Schedules vs. completed visits; geofence on-site time; photos / proof of work; inspection scores; service issues and complaints; vendor assignments; site location	Customer contract price and contracted scope; **actual scope delivered vs. contracted**; vendor invoices (real spend) and vendor quotes; site square footage and type; local labour cost (market wage) vs. actual labour cost; vendor base locations; renewal dates

The outside data is what turns Peazy's "what happened" into "what it cost and what we were paid for it".

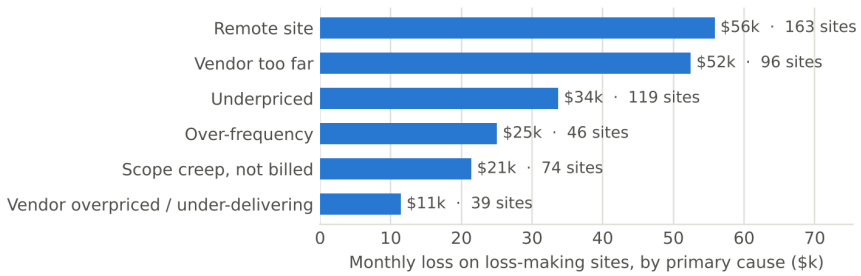
3. How I identify root causes

For each site I compute monthly margin, then compare it with a **benchmark**: what a similar site (type, size, frequency, local wage) should cost and be priced at. Rules on observable data tag each losing site with one or more of six causes and put a dollar value on each, so the biggest leak becomes the **primary cause**.

Cause	Plain meaning	Data signal	Matching fix
Underpriced	We quoted the customer too low	Price < 88% of benchmark for similar sites	Reprice at renewal from local wage x sqft x frequency
Scope creep	Customer added work, price never changed	Actual hours > 1.2x contract, frequent extra work orders	Bill change orders; lock scope in Peazy
Remote site	Site is isolated, travel dominates	Nearest other site > 30 miles	Travel surcharge, bundle into a route, or exit
Over-frequency	Cleaned more often than needed / contracted	Scheduled > contracted visits, inspection score stays high	Align schedule to real need
Vendor too far	Wrong vendor assigned, drives in from far away	Vendor travel > 25 mi while the site sits in a dense area	Reassign to the nearest zone vendor
Vendor overpriced / under-delivering	Vendor charges too much, or bills hours not worked	Rate > 1.25x metro median, or geofence time < 75% of paid hours	Pay verified hours + quality bonus; rebid

Does it work? I simulated 2,000 sites and secretly planted these six causes. Using only observable fields, the rules found them again with 95-100% precision and 88-100% recall. 27% of sites lose money ($202k/month), and 17% of losing sites have two or more causes, which is why a single fix like "cut vendor price" misses most of the loss.

Most losses come from geography and pricing, not vendor rates

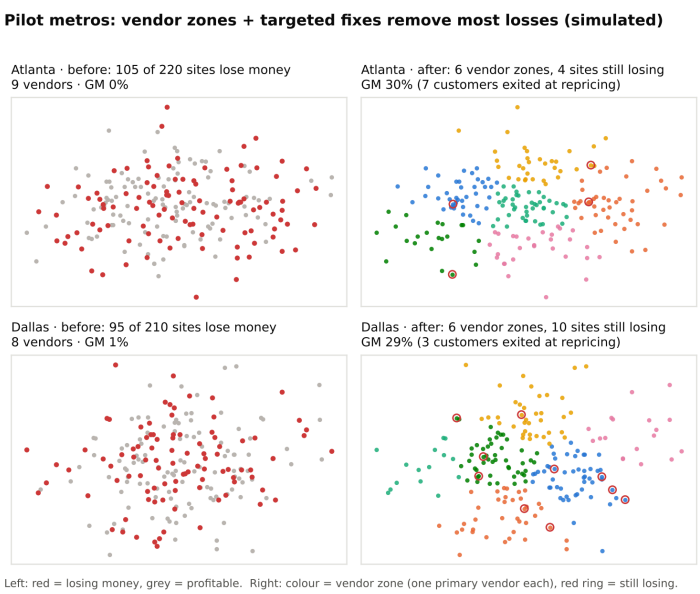


4. Solution and the role of technology

General model changes

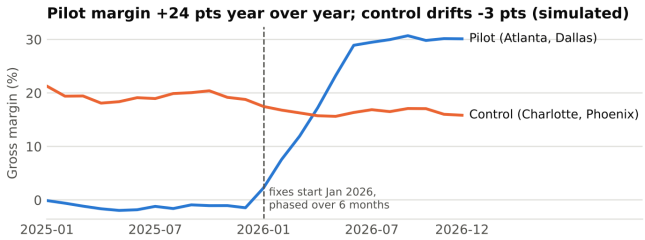
- **Assign vendors by geography:** cluster sites into zones (k-means on location) and give each zone one primary vendor with a route, so travel is shared across sites.
- **Pay for verified results:** base fee for geofence-verified hours plus a bonus tied to inspection scores and complaint-free months.
- **Set frequency by need:** use inspection and traffic data to adjust visits, up or down, with the customer.
- **Reprice from local labour cost:** quote from wage x sqft x frequency + travel; raise prices where the site cannot be made profitable otherwise.

Site-level treatment: the model outputs one row per losing site: its causes, the dollars each cause costs, and the recommended fix (*loss_locations_diagnosis.csv*). Example from Dallas: **L0345**, a 5,650 sq ft retail site, loses $2,630/month: it is underpriced, cleaned more often than contracted, and served by a vendor 25+ miles away, so it gets three fixes, not one.



Two-city example (Atlanta, Dallas): both start near 0% margin with 17 vendors between them. After zoning (12 zone vendors), repricing, change orders, schedule alignment and pay-for-verified-hours, assuming only 75% of fixes land and 15% of repriced customers leave:

Atlanta + Dallas (430 sites)	Before	After
Loss-making sites	200	14
Monthly loss on those sites	$88.9k	$2.3k
Gross margin	0.3%	29%
Monthly margin	$2k	$209k
YoY margin change	+24 pts (controls: -3 pts)	



Where technology fits: a margin + diagnosis pipeline that refreshes monthly from Peazy and invoices; a dashboard of losing sites ranked by recoverable dollars; automatic alerts when geofence time falls below paid hours or extra work orders pile up; AI photo checks to grade proof-of-work at scale; a pricing model for every new quote and renewal; zone and route optimisation when sites are added.

5. How I measure success

- **Profit:** number of loss-making sites and their share of total sq ft; expected vs. actual gross margin per site; month-over-month and year-over-year change.
- **Quality guardrails:** inspection score, photo-verified completion rate, complaints per site, customer renewal rate. These must not fall.
- **Market:** change in vendor quotes vs. market labour rates, and how much customers accept at the new price (renewal acceptance, change-order acceptance).
- **Test design:** pilot metros vs. similar control metros over the same months, so seasonality and wage inflation do not get counted as wins.

6. What I would build or test first

1. **Weeks 1-3:** join Peazy, contract and invoice data into one site-level margin table; run the six-cause diagnosis; validate 20 sites by hand with ops.
2. **Weeks 4-8:** quick wins with no customer conversation: reassign far vendors, align schedules to contract, start paying geofence-verified hours.
3. **Months 3-6:** zone two pilot metros with one vendor per zone; reprice underpriced sites and bill scope changes at renewal; compare with two control metros.

7. AI tools used

I used **Claude (Anthropic)** (1) to brainstorm and pressure-test the framework; the causes, data list, solutions and metrics are my own choices, and (2) to help write the Python simulation, diagnostic rules and charts, which I reviewed and calibrated. All numbers come from simulated data built to test the method; real gains would depend on actual contracts and markets. Code: see the GitHub link in the submission form.